

EDUCATION

MASSACHUSETTS INSTITUTE OF TECHNOLOGY (MIT)

Cambridge, MA

Master's in Financial Mathematics

2025 – Feb 2027

- **4.9/5.0 GPA | Top 1%** of applicants – received merit-based **fellowship** for academic excellence
- Teaching Assistant: Programming for Finance Professionals; Mathematical Methods for Quantitative Finance (edX)

STANFORD UNIVERSITY

Stanford, CA

Graduate Certificate in Statistics (For-Credit Graduate Coursework)

In Progress

- **3.9/4.0 GPA** | Completed **3 of 4 Stanford graduate courses** in regression, statistical learning, and stochastic processes

UNIVERSITY COLLEGE LONDON (UCL)

London, UK

Bachelor's in Economics

2022 – 2025

- **Class Rank: 2 / 430 | Top of cohort** in Years 1 and 3 | First Class Honors
- Bloomsbury Capital (student fund): led 4 analysts producing discretionary L/S stock pitches every two months

Relevant Courses: Graduate Machine Learning for Finance, Graduate Deep Learning, Graduate Optimization Methods, Quantitative Investment Management, PhD Asset Pricing, Financial Markets, Statistical Learning and Data Science, Econometrics

QUANTITATIVE RESEARCH PROJECTS

GRU LONG/SHORT EQUITY MODEL (2.36 GROSS SHARPE)

2026

- Built a **20+ year point-in-time dataset for the Russell 1000** from CRSP and MIT-accessible financial databases, engineering **daily stock-level features** across returns and price dynamics, trading activity, liquidity, volatility and risk, momentum/reversal, market and sector exposures, and firm characteristics.
- Achieved **2.36 gross Sharpe** and **21.6% gross annualized return** over the 2016–2019 development period with a **single-layer GRU** using 64 hidden units and a $64 \rightarrow 32 \rightarrow 1$ prediction head to rank next-day returns; developed internal diagnostics to identify **hidden-state transport degradation** and improve long-horizon conditioning.
- Ran a **daily long/short strategy** using the top and bottom 20% of model-ranked stocks, equal-weighted at **+100% long / -100% short**; achieved a **7.0% maximum drawdown** and **3.59 bps break-even transaction cost** per dollar traded.

INDEX REPLICATION WITH SPARSE PORTFOLIO OPTIMIZATION

2026

- Replicated **daily S&P 500 returns with a 30-stock portfolio** to study covariance estimation for portfolio construction: an optimizer chose which 30 stocks to hold and at what weights by minimizing tracking error, with only the covariance estimator varied (sample, linear shrinkage, PCA factor model, nonlinear shrinkage).
- **Nonlinear shrinkage** tracked best: **2.6% annualized tracking error**, half the 5.3% of a naive market-cap top 30, with realized risk within 5% of forecast; at 30 names it beat the factor model by 0.44 pp and came within 0.3 pp of that model at 100 names.

WELLINGTON MANAGEMENT – MIT JOINT FINANCE LAB

2025 – 2026

Action Learning Project, Fixed Income Return Prediction

- Modeled next-month excess returns of **seven global bond sectors** with a regularized regression on carry, spread, momentum, and macro signals, to guide a discretionary PM's over- and underweights against the Bloomberg Global Aggregate.
- Backtested as monthly benchmark tilts with walk-forward refits: **1.29 net information ratio** over 5 years out of sample (73% of months outperformed the benchmark); the edge came from favoring higher-carry sectors rather than timing spread moves.

EXPERIENCE

PENTA INVESTMENTS

Summer 2025

Leading European PE firm with EUR 18B AUM – healthcare, retail, media, and real estate

Investment Analyst Intern (full-time offer received)

- Sized an unregulated **gambling market** at ~\$8B with no official data: fit a panel **ridge regression** of gross gambling revenue on reporting countries and extrapolated; validated via **leave-one-country-out CV** and country-level **bootstrap intervals**.
- Conducted **financial due diligence** on an insurer with **\$75M in annual premiums**, evaluating profitability, solvency, and investment risk; benchmarked product market shares against peers to **inform** the investment committee's **acquisition decision**.

MCKINSEY & COMPANY

Summer 2024

Business Analyst Intern (full-time offer received)

- Scraped multi-year energy auction data, modeled **price distributions and bidding patterns**, and optimized power purchase agreement (PPA) vs. spot market allocation against production forecasts, projecting **+23% revenue uplift**.
- Designed a multi-criteria vendor evaluation framework for a **multi-billion-dollar energy investment**, defining how to score vendors on supply-chain and equity-partnership criteria; **adopted as the basis for the client's selection decision**.

MATH OLYMPIADS, ACADEMIC AWARDS & TECHNICAL SKILLS

- **Mathematics:** Won regional round of the Czech Math Olympiad
- **Academic Awards:** Dean's Fellow for Academic Excellence (MIT), Eleanora Allan and Hume-Lloyd prizes (Top 3, UCL)
- **Technical Skills:** Python (experience with NumPy, pandas, statsmodels, PyTorch, scikit-learn), Gurobi, Excel